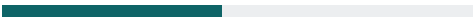


Network+

Exam N10-009

EXAM DOMAINS		WEIGHT
Networking Concepts		23%
Network Implementation		20%
Network Operations		19%
Network Security		14%
Network Troubleshooting		24%

Up to 90 questions · 90 minutes · Passing score 720 (scale 100–900)

Contents

Networking Concepts	4
1.1 The OSI Reference Model	4
1.2 Networking Appliances, Applications, and Functions	5
1.3 Cloud Concepts and Connectivity	6
1.4 Ports, Protocols, and Services	7
1.5 Transmission Media and Transceivers	8
1.6 Topologies, Architectures, and Types	10
1.7 IPv4 Network Addressing	11
1.8 Evolving Use Cases for Modern Networks	13
 Network Implementation	 15
2.1 Routing Technologies	15
2.2 Switching Technologies	16
2.3 Wireless Devices and Technologies	17
2.4 Physical Installations	19
 Network Operations	 21
3.1 Organizational Processes and Procedures	21
3.2 Network Monitoring Technologies	22
3.3 Disaster Recovery and High Availability	23
3.4 Network Services	23
3.5 Network Access and Management Methods	25
 Network Security	 26
4.1 Basic Network Security Concepts	26
4.2 Types of Attacks	27
4.3 Network Security Features, Defense Techniques, and Solutions	29
 Network Troubleshooting	 30
5.1 The Troubleshooting Methodology	30

5.2 Cabling and Physical Interface Issues	31
5.3 Network Service Issues	32
5.4 Performance Issues	32
5.5 Tools and Protocols for Troubleshooting	33
 Practice Questions	 35
 Answer Key	 41

01

Networking Concepts

DOMAIN 1 • 23% OF THE EXAM

This domain is the foundation for everything else. If the OSI model, ports, and subnetting feel solid, the rest of the exam gets much easier. Put extra time into 1.4 (ports) and 1.7 (IPv4 addressing), because both show up everywhere, including in the performance-based questions.

1.1 The OSI Reference Model

The OSI model splits network communication into seven layers. Each layer does one job and hands its data to the layer above or below it. On the exam you'll mostly use it two ways: identifying which layer a device or protocol works at, and troubleshooting bottom-up or top-down.

#	LAYER	WHAT IT DOES	DATA UNIT (PDU)	EXAMPLES
7	Application	Interface between network and user applications	Data	HTTP, DNS, SMTP, FTP, SSH
6	Presentation	Formatting, encryption, compression	Data	TLS/SSL, JPEG, ASCII
5	Session	Opens, manages, and closes sessions	Data	RPC, NetBIOS, SQL sessions
4	Transport	End-to-end delivery, ports, reliability	Segment (TCP) / Datagram (UDP)	TCP, UDP
3	Network	Logical addressing and routing between networks	Packet	IP, ICMP, routers, L3 switches
2	Data Link	MAC addressing and local delivery, error detection (FCS)	Frame	Ethernet, switches, ARP, 802.1Q
1	Physical	Bits on the wire, radio, or light	Bit	Cables, hubs, repeaters, transceivers

REMEMBER

Mnemonic, top to bottom: **All People Seem To Need Data Processing**. Bottom to top: **Please Do Not Throw Sausage Pizza Away**.

Encapsulation is the process of each layer wrapping the data from the layer above with its own header as it moves down the stack (data → segment → packet → frame → bits). **De-encapsulation** is the reverse on the receiving side. Layer 2 also adds a trailer, the Frame Check Sequence (FCS),

which is used to detect corrupted frames.

The **TCP/IP model** is the practical four-layer version: Application (OSI 5–7), Transport (4), Internet (3), and Network Access/Link (1–2).

1.2 Networking Appliances, Applications, and Functions

PHYSICAL AND VIRTUAL APPLIANCES

APPLIANCE	OSI LAYER	PURPOSE
Router	3	Forwards packets between different networks using IP addresses and a routing table. Separates broadcast domains.
Switch	2 (L3 switches also route)	Forwards frames within a LAN using a MAC address table. Each port is its own collision domain.
Firewall	3–7	Permits or denies traffic based on rules. Next-generation firewalls (NGFW) inspect applications and content, not just ports.
IDS	3–7	Intrusion Detection System. Monitors and alerts on suspicious traffic. Passive, usually fed by a tap or SPAN port.
IPS	3–7	Intrusion Prevention System. Sits inline and can block malicious traffic in real time.
Load balancer	4–7	Spreads incoming requests across multiple servers for performance and availability.
Proxy	7	Makes requests on behalf of clients. Used for filtering, caching, and hiding internal clients. A reverse proxy sits in front of servers.
NAS	Storage	Network Attached Storage. File-level storage over the LAN (SMB, NFS).
SAN	Storage	Storage Area Network. Block-level storage on a dedicated network (Fibre Channel, iSCSI). Servers see it like a local disk.
Wireless AP	2	Bridges wireless clients onto the wired network.
Wireless LAN controller	2–3	Centrally manages many lightweight APs: config, channels, power, roaming.

REMEMBER

IDS = detect and alert (passive). IPS = detect and block (inline). If a question says the device “dropped” or “stopped” an attack, it’s an IPS.

APPLICATIONS AND FUNCTIONS

- **CDN (Content Delivery Network)**: geographically distributed servers that cache content close to users to cut latency and load on the origin server.

- **VPN (Virtual Private Network):** an encrypted tunnel across an untrusted network such as the internet.
- **QoS (Quality of Service):** prioritizes sensitive traffic such as voice and video so it isn't delayed by bulk traffic.
- **TTL (Time to Live):** a hop counter in the IP header. Each router decrements it by 1, and the packet is dropped at 0, which prevents routing loops from circulating packets forever. DNS records also have a TTL, which controls how long they're cached.

1.3 Cloud Concepts and Connectivity

SERVICE MODELS

MODEL	PROVIDER MANAGES	YOU MANAGE	EXAMPLE
SaaS	Everything	Your data and users	Microsoft 365, Gmail
PaaS	Hardware, OS, runtime	Your application and data	Heroku, Azure App Service
IaaS	Hardware, virtualization, networking	OS, apps, data	AWS EC2, Azure VMs

DEPLOYMENT MODELS

- **Public:** shared provider infrastructure (AWS, Azure, GCP).
- **Private:** cloud infrastructure dedicated to one organization, either on-premises or hosted.
- **Hybrid:** a mix of public and private, connected so workloads can move between them.

KEY CLOUD NETWORKING TERMS

- **NFV (Network Functions Virtualization):** running network appliances (firewalls, routers, load balancers) as software VMs instead of dedicated hardware.
- **VPC (Virtual Private Cloud):** your own logically isolated network inside a public cloud.
- **Network security groups:** stateful, firewall-like rules applied to instances or interfaces.
- **Network security lists:** rules applied at the subnet level (stateless in some providers).
- **Internet gateway:** lets resources in a VPC talk to and be reached from the internet.
- **NAT gateway:** lets private resources reach the internet outbound without being reachable inbound.
- **Cloud connectivity options:** a VPN over the internet (cheaper, quick to set up) or a **direct connect** private dedicated link (consistent performance, more expensive).
- **Scalability** is the ability to add resources to handle growth. **Elasticity** is scaling up and down automatically with demand.
- **Multitenancy:** multiple customers share the same physical infrastructure, logically isolated from one another.

1.4 Ports, Protocols, and Services

Memorize this table. Expect direct questions (“which port does X use?”) and scenario questions where you have to pick the right firewall rule.

PORT(S)	PROTOCOL	TCP/UDP	NOTES
20, 21	FTP	TCP	21 = control, 20 = data (active mode). Unencrypted.
22	SSH / SFTP	TCP	Secure remote shell. SFTP runs over SSH.
23	Telnet	TCP	Unencrypted remote shell. Replace with SSH.
25	SMTP	TCP	Sending mail between servers.
53	DNS	UDP and TCP	UDP for queries, TCP for zone transfers and large responses.
67, 68	DHCP	UDP	67 = server, 68 = client.
69	TFTP	UDP	Trivial FTP. No authentication. Used for config and firmware files.
80	HTTP	TCP	Unencrypted web.
123	NTP	UDP	Time synchronization.
161, 162	SNMP	UDP	161 = queries to agents, 162 = traps sent to the manager.
389	LDAP	TCP/UDP	Directory services (e.g., Active Directory).
443	HTTPS	TCP	HTTP over TLS.
445	SMB	TCP	Windows file and printer sharing.
514	Syslog	UDP	Log forwarding to a collector.
587	SMTP (submission, secure)	TCP	Client mail submission with STARTTLS. Listed as SMTPS in the objectives.
636	LDAPS	TCP	LDAP over TLS.
1433	Microsoft SQL Server	TCP	Database.
3389	RDP	TCP	Remote Desktop Protocol.
5060, 5061	SIP	TCP/UDP	VoIP call setup. 5061 = SIP over TLS.

REMEMBER

Secure versions pair up nicely: Telnet 23 → SSH 22, HTTP 80 → HTTPS 443, LDAP 389 → LDAPS 636, SIP 5060 → 5061, FTP 21 → SFTP 22.

IP PROTOCOL TYPES

- **TCP**: connection-oriented and reliable. Uses the three-way handshake (SYN → SYN-ACK → ACK), sequencing, acknowledgments, and retransmission. Use it when accuracy matters (web, email, file transfer).
- **UDP**: connectionless, no guarantees, low overhead. Use it when speed matters more than perfection (voice, video, DNS queries, DHCP).
- **ICMP**: error and diagnostic messages. Used by ping (echo request/reply) and traceroute.
- **GRE (Generic Routing Encapsulation)**: tunnels other protocols inside IP. No encryption on its own.
- **IPsec**: encrypts and authenticates IP traffic. **AH** (Authentication Header) provides integrity and authentication but *no encryption*. **ESP** (Encapsulating Security Payload) adds encryption. **IKE** (Internet Key Exchange) negotiates the keys and security associations.

TRAFFIC TYPES

- **Unicast**: one sender to one receiver.
- **Broadcast**: one to all hosts on the local subnet (IPv4 only, e.g., 255.255.255.255). Routers don't forward broadcasts.
- **Multicast**: one to a subscribed group (224.0.0.0–239.255.255.255 in IPv4).
- **Anycast**: one to the *nearest* of several hosts sharing the same address. Common with DNS root servers and CDNs.

REMEMBER

IPv6 has no broadcast. It uses multicast instead.

1.5 Transmission Media and Transceivers

WIRELESS AND OTHER MEDIA

- **802.11 (Wi-Fi)**: covered in depth in 2.3.
- **Cellular**: 4G LTE and 5G. Used for WAN backup links and remote sites.
- **Satellite**: reaches remote areas but has high latency with geostationary satellites. Low-earth-orbit (LEO) services like Starlink have much lower latency.

COPPER (802.3 ETHERNET) CABLE CATEGORIES

CATEGORY	MAX SPEED	MAX DISTANCE	NOTES
Cat 5	100 Mbps	100 m	Legacy
Cat 5e	1 Gbps	100 m	Common minimum today
Cat 6	1 Gbps (10 Gbps ≤ 55 m)	100 m	
Cat 6a	10 Gbps	100 m	Augmented, better shielding against crosstalk
Cat 7	10 Gbps	100 m	Shielded
Cat 8	25/40 Gbps	30 m	Data center, short runs

- **STP vs. UTP:** shielded twisted pair resists EMI (electromagnetic interference). Unshielded is cheaper and most common.
- **Plenum** cable has a fire-resistant jacket that gives off less toxic smoke. It's required in air-handling spaces (above drop ceilings, under raised floors). **Non-plenum (PVC)** is cheaper but not allowed there.
- **Coaxial:** used for cable internet (RG-6 with F-type connectors).
- **DAC / Twinax:** Direct Attach Copper, a short copper cable with transceivers built onto each end. A cheap, low-latency way to link switches and servers in the same rack.
- **Wiring standards:** T568A and T568B. Same standard on both ends = straight-through cable. A on one end and B on the other = crossover. (Modern ports use Auto-MDIX, so crossovers are rarely needed.)

PIN	1	2	3	4	5	6	7	8
T568B	White/O range	Orange	White/G reen	Blue	White/Bl ue	Green	White/B rown	Brown
T568A	White/G reen	Green	White/O range	Blue	White/Bl ue	Orange	White/B rown	Brown

FIBER OPTIC

TYPE	CORE	LIGHT SOURCE	DISTANCE	COST
Single-mode (SMF)	Thin (~9 μm)	Laser	Long (kilometers)	Higher
Multimode (MMF)	Wider (50/62.5 μm)	LED or VCSEL	Short (hundreds of meters)	Lower

Fiber is immune to EMI and hard to tap, which is why it's used for backbones and long runs. Common standards: **1000BASE-SX** / **10GBASE-SR** (short reach, multimode) and **1000BASE-LX** / **10GBASE-LR** (long reach, single-mode).

TRANSCEIVERS AND CONNECTORS

- **Transceivers** plug into switch slots and convert electrical signals to light or another medium. **SFP** handles 1 Gbps, **SFP+** 10 Gbps, **QSFP/QSFP+** 40 Gbps (four lanes), and **QSFP28** 100 Gbps. Protocols carried include Ethernet and Fibre Channel.
- **Both ends must match** in speed, wavelength, and fiber type. A mismatch is a classic troubleshooting answer.

CONNECTOR	USED WITH	HOW TO RECOGNIZE IT
LC	Fiber	Small form factor, push-and-latch. Most common on SFPs.
SC	Fiber	Square, push-pull.
ST	Fiber	Round bayonet, push and twist (“stick and twist”).
MPO	Fiber	Single connector carrying many fibers (12 or 24). High density, used with QSFP.
RJ45	Copper Ethernet	8-pin modular plug.
RJ11	Telephone / DSL	Smaller, 2–6 pins.
F-type	Coax	Screw-on. Cable modems and TV.
BNC	Coax	Bayonet twist-lock. Legacy and video.

1.6 Topologies, Architectures, and Types

TOPOLOGY	DESCRIPTION	PROS / CONS
Star / hub-and-spoke	All devices connect to a central switch or hub. In WANs, branches connect to a central site.	Easy to manage and troubleshoot. The center is a single point of failure.
Mesh	Devices interconnect with multiple paths. Full mesh = every node links to every other.	Highly redundant but expensive. Full mesh needs $n(n-1)/2$ links.
Point-to-point	A direct link between two nodes.	Simple and dedicated. Doesn’t scale.
Hybrid	A combination of the above.	Flexible. More complex.
Spine-and-leaf	Data center design. Every leaf (access) switch connects to every spine switch.	Predictable low latency: every server is the same number of hops apart. Great for east-west traffic.

HIERARCHICAL ARCHITECTURES

- **Three-tier:** **Core** (fast backbone, no filtering), **Distribution** (routing, policy, ACLs, aggregates access switches), and **Access** (where end devices plug in).
- **Collapsed core:** core and distribution combined into one layer. Common in smaller networks to save cost.

TRAFFIC FLOWS

- **North-south:** traffic entering or leaving the data center (client to server, internet).
- **East-west:** traffic between servers inside the data center. It's now the majority of data center traffic, which is why spine-and-leaf became popular.

NETWORK TYPES (SIZES)

PAN (personal, e.g., Bluetooth) → LAN (one building) → CAN (campus) → MAN (city) → WAN (wide area, e.g., the internet or links between offices). Also **WLAN** (wireless LAN) and **SAN** (storage network).

1.7 IPv4 Network Addressing

An IPv4 address is 32 bits, written as four octets (e.g., 192.168.10.25). The **subnet mask** marks which bits are the network portion and which are the host portion. **CIDR** notation writes the mask as the number of network bits: 255.255.255.0 = /24.

SPECIAL ADDRESS RANGES

RANGE	PURPOSE
10.0.0.0/8	Private (RFC 1918)
172.16.0.0/12 (172.16.0.0 – 172.31.255.255)	Private (RFC 1918)
192.168.0.0/16	Private (RFC 1918)
127.0.0.0/8	Loopback. 127.0.0.1 tests your own TCP/IP stack.
169.254.0.0/16	APIPA. Self-assigned when a host can't reach DHCP. A big troubleshooting clue.
100.64.0.0/10	Carrier-grade NAT (shared address space)

REMEMBER

If a client shows a 169.254.x.x address, the problem is DHCP: the server is down, the scope is exhausted, there's no DHCP relay, or the client is on the wrong VLAN or has a bad cable.

CLASSFUL ADDRESSING (LEGACY, STILL TESTED)

CLASS	FIRST OCTET	DEFAULT MASK	USE
A	1–126	/8 (255.0.0.0)	Large networks
B	128–191	/16 (255.255.0.0)	Medium networks
C	192–223	/24 (255.255.255.0)	Small networks
D	224–239	n/a	Multicast
E	240–255	n/a	Experimental

SUBNETTING: THE METHOD

- **Hosts per subnet** = $2^h - 2$, where h is the number of host bits (subtract the network and broadcast addresses).
- **Number of subnets** = 2^s , where s is the number of bits borrowed from the host portion.
- **Block size (magic number)** = 256 – the mask value in the “interesting” octet (the one that isn’t 255 or 0). Subnets start at multiples of the block size.
- **Network address** = the first address in the block. **Broadcast** = the last address. **Usable hosts** are everything in between.

CIDR	MASK	BLOCK SIZE	USABLE HOSTS
/24	255.255.255.0	256	254
/25	255.255.255.128	128	126
/26	255.255.255.192	64	62
/27	255.255.255.224	32	30
/28	255.255.255.240	16	14
/29	255.255.255.248	8	6
/30	255.255.255.252	4	2 (point-to-point links)
/31	255.255.255.254	2	2 (point-to-point, RFC 3021)
/32	255.255.255.255	1	1 (single host route)

WORKED EXAMPLE

Question: What are the network address, broadcast address, and usable range for 192.168.10.77/27?

- /27 means the mask is 255.255.255.224, so the block size is $256 - 224 = 32$.
- Count up in 32s in the last octet: 0, 32, **64**, 96 ... 77 falls in the block starting at 64.
- Network = **192.168.10.64**. Broadcast = one less than the next block = **192.168.10.95**.
- Usable range = **192.168.10.65 – 192.168.10.94** (30 hosts).

VLSM (Variable Length Subnet Masking) means using different mask lengths within the same address space, so each subnet is sized to what it needs: a /30 for a router link, a /24 for a user floor, and so on. Allocate the largest subnets first to avoid overlap.

REMEMBER

Practice subnetting until you can do a /26–/30 problem in under a minute. It’s one of the highest-value skills on the exam.

1.8 Evolving Use Cases for Modern Networks

- **SDN (Software-Defined Networking)**: separates the **control plane** (decides where traffic goes) from the **data plane** (actually forwards it), with a central controller programming the devices. The **management plane** is how admins configure devices. SDN controllers talk to devices through “southbound” APIs and to apps through “northbound” APIs.
- **SD-WAN**: SDN applied to WAN links. Key features: **application-aware** routing (sends voice over the best link), **zero-touch provisioning** (devices configure themselves when plugged in), **transport agnostic** (works over broadband, MPLS, LTE, anything), and **central policy management**.
- **VXLAN**: encapsulates Layer 2 frames in UDP so Layer 2 networks can stretch across Layer 3 networks. Used for **data center interconnect (DCI)**. It supports about 16 million segments versus 4,094 VLANs.
- **Zero Trust Architecture (ZTA)**: “never trust, always verify.” No device or user is trusted just because it’s inside the network. Relies on policy-based authentication, explicit authorization for every request, and least-privilege access.
- **SASE / SSE**: Secure Access Service Edge combines SD-WAN with cloud-delivered security. Security Service Edge (SSE) is the security half: secure web gateway, CASB, and zero-trust network access.
- **Infrastructure as Code (IaC)**: defining network and server config in files instead of by hand. Uses **playbooks**, **templates**, and **reusable tasks** (e.g., Ansible, Terraform). Prevents **configuration drift**, helps compliance, and makes upgrades repeatable. **Dynamic inventories** automatically discover what to manage.
- **Source control**: version control (e.g., Git) for configs and code. Gives you a central repository, history, branching, and conflict identification when two people change the same thing.

IPv6

IPv6 uses 128-bit addresses written as eight groups of hex (e.g., 2001:0db8:0000:0000:0000:ff00:0042:8329). Shorten them by dropping leading zeros in each group and replacing **one** run of all-zero groups with :: → 2001:db8::ff00:42:8329. Its main purpose is **mitigating IPv4 address exhaustion**.

ADDRESS / PREFIX	MEANING
::1	Loopback
fe80::/10	Link-local. Auto-configured on every IPv6 interface, not routable.
fc00::/7	Unique local (the IPv6 equivalent of private addresses)
2000::/3	Global unicast (public, routable)
ff00::/8	Multicast

- **Compatibility methods**: **dual stack** (run IPv4 and IPv6 side by side, the most common approach), **tunneling** (carry IPv6 inside IPv4, e.g., 6to4), and **NAT64** (translates between IPv6-only clients

and IPv4 servers).

- **Neighbor Discovery Protocol (NDP)** replaces ARP in IPv6. **SLAAC** lets hosts configure their own addresses (covered in 3.4).

02

Network Implementation

DOMAIN 2 · 20% OF THE EXAM

This domain is about actually building the network: routing, switching, wireless, and the physical install. Expect performance-based questions here, such as configuring a VLAN, placing APs, or picking the right routing choice.

2.1 Routing Technologies

STATIC VS. DYNAMIC ROUTING

- **Static routes** are entered by hand. They're simple, secure, and use no bandwidth, but they don't adapt to failures and don't scale. A **default route** (0.0.0.0/0) is a static route used when nothing more specific matches, often called the gateway of last resort.
- **Dynamic routing protocols** let routers share routes automatically and adapt when links fail.

PROTOCOL	TYPE	METRIC	WHERE IT'S USED
OSPF	Link-state (interior)	Cost (based on bandwidth)	Enterprise networks. Open standard. Organized into areas, with Area 0 as the backbone. Fast convergence.
EIGRP	Advanced distance-vector / hybrid (interior)	Bandwidth and delay by default	Originally Cisco proprietary. Fast convergence.
BGP	Path-vector (exterior)	Path attributes (e.g., AS path)	Routing between autonomous systems. The routing protocol of the internet.
RIP	Distance-vector (interior)	Hop count (max 15)	Legacy. Slow convergence. Not in N10-009 objectives but still good to recognize.

ROUTE SELECTION: HOW A ROUTER CHOOSES

A router decides in this order:

- **1. Longest prefix match (prefix length):** the most specific route always wins. A /28 beats a /24 beats a /0, regardless of protocol.
- **2. Administrative distance (AD):** if two *different sources* offer the same prefix, the lower AD is more trusted.
- **3. Metric:** if the same protocol offers multiple paths to the same prefix, the lowest metric wins.

ROUTE SOURCE	DEFAULT AD
Directly connected	0
Static	1
eBGP	20
EIGRP (internal)	90
OSPF	110
RIP	120
iBGP	200

ADDRESS TRANSLATION AND REDUNDANCY

- **NAT (Network Address Translation):** translates private addresses to public addresses at the network edge. **Static NAT** is one-to-one (useful for exposing an internal server).
- **PAT (Port Address Translation, or NAT overload):** many private hosts share one public IP, distinguished by port number. This is what home routers do.
- **FHRP (First Hop Redundancy Protocol):** two or more routers share a **virtual IP (VIP)** that hosts use as their default gateway. If the active router fails, a standby takes over the VIP, so hosts never notice. Examples: **HSRP** (Cisco), **VRRP** (open standard), and **GLBP**.
- **Subinterfaces:** one physical router interface split into logical interfaces, each tagged for a different VLAN. This is “router on a stick” inter-VLAN routing.

2.2 Switching Technologies

VLANS

A **VLAN** splits one physical switch into multiple logical broadcast domains. That improves security (separates traffic), performance (smaller broadcast domains), and organization (group by function, not location). Traffic between VLANs must be routed, either by a router or a Layer 3 switch.

- **VLAN database:** where the switch stores its VLAN definitions (IDs 1–4094).
- **SVI (Switch Virtual Interface):** a virtual Layer 3 interface for a VLAN on a Layer 3 switch. Give it an IP and it becomes the default gateway for that VLAN.
- **Access port:** carries one VLAN, untagged. For end devices.
- **Trunk port:** carries many VLANs between switches (or to a router/server), using **802.1Q tagging**, which inserts a 4-byte tag holding the VLAN ID into the frame.
- **Native VLAN:** the one VLAN on a trunk whose traffic is sent *untagged*. Default is VLAN 1. It must match on both ends, and changing it from VLAN 1 is a security best practice (it helps prevent VLAN hopping).
- **Voice VLAN:** a separate VLAN for IP phones, so voice traffic can be prioritized with QoS. A phone and a PC can share one switch port, with the phone on the voice VLAN and the PC on the data

VLAN.

INTERFACE CONFIGURATION

- **Link aggregation (LACP, 802.3ad)**: bundles multiple physical links into one logical link for more bandwidth and redundancy. Also called port channel, EtherChannel, or NIC teaming.
- **Speed and duplex**: both ends must match. A **duplex mismatch** (one side full, one side half) causes slow performance, late collisions, and CRC errors. Autonegotiation on both sides is the safe default.
- **MTU (Maximum Transmission Unit)**: the largest frame payload, 1500 bytes by default for Ethernet. **Jumbo frames** (up to about 9000 bytes) cut overhead for storage and data center traffic, but *every device in the path* must support them, or you'll get drops and fragmentation.

SPANNING TREE PROTOCOL (STP, 802.1D)

Redundant links between switches create **Layer 2 loops**, which cause **broadcast storms** that can take down a network in seconds (Ethernet frames have no TTL). STP prevents this by blocking redundant paths and unblocking them only if the active path fails.

- Switches elect a **root bridge**: the switch with the **lowest bridge ID** (priority, default 32768, plus MAC address). Lower priority wins. On a tie, the lowest MAC wins.
- **Port roles**: **root port** (each non-root switch's best path toward the root), **designated port** (the forwarding port on each segment), and **blocked / alternate port** (kept on standby to prevent loops).
- **Port states (802.1D)**: Blocking → Listening → Learning → Forwarding (plus Disabled). Classic STP takes 30–50 seconds to converge. **RSTP (802.1w)** converges in a few seconds.
- **PortFast** skips the listening/learning states on ports connected to end devices. **BPDUGuard** shuts a PortFast port down if a switch is plugged into it.

REMEMBER

Always set the root bridge priority manually on your core switch. Otherwise, an old switch with a low MAC address might win the election and push traffic through a slow path.

2.3 Wireless Devices and Technologies

802.11 STANDARDS

STANDARD	WI-FI NAME	FREQUENCY	MAX SPEED (THEORETICAL)
802.11a	–	5 GHz	54 Mbps
802.11b	–	2.4 GHz	11 Mbps
802.11g	–	2.4 GHz	54 Mbps
802.11n	Wi-Fi 4	2.4 and 5 GHz	600 Mbps (MIMO)
802.11ac	Wi-Fi 5	5 GHz	~6.9 Gbps (MU-MIMO downlink)
802.11ax	Wi-Fi 6 / 6E	2.4, 5 (and 6 GHz for 6E)	~9.6 Gbps (OFDMA)
802.11be	Wi-Fi 7	2.4, 5, 6 GHz	~46 Gbps

FREQUENCIES AND CHANNELS

- **2.4 GHz:** longer range and better wall penetration, but slower and crowded. Only **three non-overlapping channels: 1, 6, and 11** (in North America). Also shares spectrum with microwaves and Bluetooth.
- **5 GHz:** faster with many more non-overlapping channels, but shorter range.
- **6 GHz (Wi-Fi 6E/7):** the most clean spectrum and least interference, but the shortest range.
- **Channel width:** 20, 40, 80, or 160 MHz (320 MHz with Wi-Fi 7). Wider channels mean more throughput but fewer non-overlapping channels and more interference. Stick to 20 MHz on 2.4 GHz.
- **Band steering:** the AP nudges dual-band clients onto the less crowded 5 or 6 GHz band.
- **Regulatory impacts:** allowed channels and power vary by country. **802.11h** adds DFS (Dynamic Frequency Selection, avoids radar on some 5 GHz channels) and TPC (Transmit Power Control).

IDENTIFIERS AND NETWORK TYPES

- **SSID:** the network name. **BSSID:** the MAC address of one specific AP radio. **ESSID:** one SSID shared across multiple APs so clients can roam.
- **Infrastructure mode:** clients connect through an AP (the normal setup). **Ad hoc:** devices connect directly with no AP. **Mesh:** APs connect to each other wirelessly to extend coverage. **Point-to-point:** a wireless bridge linking two sites.

SECURITY AND AUTHENTICATION

OPTION	DETAILS
WPA2	AES encryption via CCMP. Personal mode uses a pre-shared key (PSK), vulnerable to offline dictionary attacks against weak passphrases.
WPA3	Current standard. Personal mode uses SAE (Simultaneous Authentication of Equals), which resists offline password cracking. Enterprise offers 192-bit mode.
PSK (Personal)	One shared password for everyone. Fine for homes.
Enterprise	Each user authenticates individually with 802.1X against a RADIUS server. Use this in businesses.
Guest network + captive portal	An isolated network where users must accept terms or log in through a web page before they get access.

REMEMBER

WEP and WPA (TKIP) are broken. If they show up as an answer choice for securing a network, they're wrong.

ANTENNAS AND AP TYPES

- **Omnidirectional**: radiates 360° horizontally. Good for general coverage from the middle of a room.
- **Directional** (Yagi, parabolic, patch): focuses the signal in one direction. Good for long point-to-point links or covering a hallway.
- **Autonomous (fat) AP**: configured and managed individually.
- **Lightweight (thin) AP**: managed centrally by a **wireless LAN controller**. Scales far better.

2.4 Physical Installations

- **MDF (Main Distribution Frame)**: the main wiring closet where the building's network connects to the outside world and the core. **IDF (Intermediate Distribution Frame)**: smaller closets, usually one per floor, that connect back to the MDF.
- **Racks**: sized in rack units (1U = 1.75 inches). Common full-height racks are 42U. Mind **port-side exhaust vs. intake** airflow so hot air from one device isn't blown into another (hot aisle / cold aisle). Use **lockable** racks for physical security.
- **Cabling**: **patch panels** terminate horizontal copper runs. **Fiber distribution panels** do the same for fiber.
- **Power**: a **UPS** (Uninterruptible Power Supply) provides battery backup for short outages and conditions power. A **PDU** (Power Distribution Unit) distributes power to rack devices. Calculate the **power load** so circuits aren't overloaded, and match the **voltage** (e.g., 120 V vs. 208/240 V). Generators handle long outages.
- **Environment**: control **temperature** (about 64–81°F / 18–27°C per ASHRAE) and **humidity** (too low causes static discharge, too high causes condensation and corrosion). **Fire suppression**

should be a clean agent (gas) system in server rooms, not water sprinklers, where possible.

03

Network Operations

DOMAIN 3 · 19% OF THE EXAM

This domain covers running the network day to day: documentation, monitoring, disaster recovery, core services like DHCP and DNS, and secure remote management.

3.1 Organizational Processes and Procedures

DOCUMENTATION

- **Physical diagram:** shows where devices and cables actually are (racks, rooms, ports). **Logical diagram:** shows how data flows: subnets, VLANs, routing, IP addressing.
- **Network diagrams by layer:** Layer 1 (physical cabling), Layer 2 (switches and VLANs), Layer 3 (routing and IP subnets).
- **Rack diagram:** what's mounted in each rack unit. **Cable map:** which cable goes where (labeled on both ends).
- **Asset inventory:** hardware, software, licensing, and warranty/support details for everything you own.
- **IPAM (IP Address Management):** tracks IP address allocation, DHCP, and DNS in one place.
- **SLA (Service Level Agreement):** the provider's commitment for uptime and response time, e.g., 99.99% availability.
- **Wireless survey / heat map:** a visual map of signal strength across a site. Used to plan AP placement and find dead zones.

LIFE CYCLE, CHANGE, AND CONFIGURATION MANAGEMENT

- **End of life (EOL):** the vendor stops selling the product. **End of support (EOS):** the vendor stops patching it. Running EOS gear is a security risk.
- **Software management:** keep patches, OS versions, and firmware up to date, tested first in a non-production environment.
- **Decommissioning:** securely wipe or destroy storage, remove the device from inventory, monitoring, and DNS.
- **Change management:** a formal request process (submit, review, approve, schedule, implement, document) with a rollback plan. It prevents outages caused by unplanned changes.
- **Configuration management:** keep the **production config**, a **backup config**, and a **baseline / golden config** (the approved standard every device should match). Comparing against the baseline detects unauthorized changes.

3.2 Network Monitoring Technologies

SNMP (SIMPLE NETWORK MANAGEMENT PROTOCOL)

- A **manager** (monitoring server) polls **agents** (on devices) on UDP 161. Agents send unsolicited **traps** (alerts) to the manager on UDP 162.
- **MIB (Management Information Base)**: the database of values a device exposes, each identified by an **OID** (object identifier).
- **Versions**: **v1/v2c** use plaintext **community strings** as passwords (“public” for read, “private” for read-write by default, which you must change). **v3** adds proper **authentication and encryption**. Always prefer v3.

OTHER MONITORING METHODS

- **Flow data** (NetFlow, sFlow, IPFIX): metadata about conversations (who talked to whom, ports, how much). Great for seeing *what* is using bandwidth without capturing every packet.
- **Packet capture**: records full packets for deep analysis (Wireshark, tcpdump).
- **Port mirroring (SPAN)**: a switch copies traffic from one or more ports to a monitoring port for an IDS or packet capture.
- **Baseline metrics**: record what “normal” looks like so **anomaly alerting** can flag deviations.
- **Log aggregation**: a **syslog collector** centralizes logs from devices. A **SIEM** (Security Information and Event Management) collects and correlates logs from everything and alerts on suspicious patterns.
- **API integration**: pulling data from devices and cloud services programmatically.

SYSLOG SEVERITY	NAME	SYSLOG SEVERITY	NAME
0	Emergency	4	Warning
1	Alert	5	Notice
2	Critical	6	Informational
3	Error	7	Debug

REMEMBER

Lower syslog number = more severe. Mnemonic: **Every Awesome Cisco Engineer Will Need Ice cream Daily**.

MONITORING SOLUTIONS

Network discovery (ad hoc or scheduled scans that find devices), traffic analysis, performance monitoring, availability monitoring (is it up?), and configuration monitoring (has the config changed?).

3.3 Disaster Recovery and High Availability

METRIC	MEANING	EXAMPLE
RPO (Recovery Point Objective)	Maximum acceptable data loss , measured in time. Drives backup frequency.	RPO 1 hour → back up at least hourly.
RTO (Recovery Time Objective)	Maximum acceptable downtime before service must be restored.	RTO 4 hours → be back up within 4 hours.
MTTR (Mean Time to Repair)	Average time to fix a failed component.	Lower is better.
MTBF (Mean Time Between Failures)	Average uptime between failures. Measures reliability.	Higher is better.

REMEMBER

Point = data (how much you can lose). Time = downtime (how long you can be down).

DR SITES

SITE	WHAT'S THERE	RECOVERY SPEED	COST
Cold	Space, power, and connectivity only. No equipment or data.	Slowest (days to weeks)	Lowest
Warm	Some equipment, partially configured. Data may be somewhat old.	Hours to days	Medium
Hot	Fully equipped mirror with near-real-time data.	Minutes	Highest

- **Active-active:** all systems handle traffic at once. Better resource use, and the load shifts if one fails.
- **Active-passive:** one system works while a standby waits to take over.
- **Testing:** **tabletop exercises** (walk through the plan in a meeting) and **validation tests** (actually fail over to prove it works).

3.4 Network Services

DHCP

DHCP automatically gives hosts an IP address, subnet mask, default gateway, DNS servers, and more. The process is **DORA**: **D**iscover (client broadcast) → **O**ffer (server) → **R**quest (client) → **A**cknowledge (server).

- **Scope:** the pool of addresses a server hands out for a subnet. **Exclusions:** addresses inside the scope that DHCP won't assign (e.g., for statically addressed printers).
- **Reservation:** always gives the same IP to a specific device, matched by MAC address.
- **Lease time:** how long a client keeps its address. Clients try to renew at 50% of the lease.

- **Options:** extra settings such as default gateway (option 3), DNS servers (option 6), and domain name.
- **DHCP relay / IP helper:** DHCP Discover is a broadcast and routers don't forward broadcasts, so a relay agent on the router forwards requests to a DHCP server on another subnet. **Missing relay = clients on that subnet get APIPA addresses.**
- **SLAAC (Stateless Address Autoconfiguration):** IPv6 hosts build their own address from the network prefix in router advertisements, with no DHCP server needed.

DNS

DNS translates names into IP addresses. A client asks a **recursive** resolver, which does the legwork of querying root, TLD, and **authoritative** servers on the client's behalf.

RECORD	PURPOSE
A	Name → IPv4 address
AAAA	Name → IPv6 address
CNAME	Alias pointing to another name (e.g., www → server1)
MX	Mail server for the domain
NS	Authoritative name servers for the zone
PTR	Reverse lookup: IP → name
TXT	Arbitrary text. Used for SPF, DKIM, and domain verification.
SOA	Start of Authority: zone settings and primary server
SRV	Location of a service (e.g., LDAP or SIP servers)

- **Forward zone:** names to IPs. **Reverse zone:** IPs to names (PTR records).
- **Authoritative** answers come from the server that owns the zone. **Non-authoritative** answers come from a cache.
- **Primary** server holds the writable copy of the zone. **Secondary** servers hold read-only copies received through zone transfers (TCP 53).
- **DNSSEC:** digitally signs DNS records so clients can verify they're authentic. Protects against DNS poisoning. It does *not* encrypt queries.
- **DoH (DNS over HTTPS, port 443)** and **DoT (DNS over TLS, port 853)** encrypt DNS queries for privacy.
- **Hosts file:** a local file mapping names to IPs that's checked *before* DNS. A malware favorite for redirecting traffic.

TIME PROTOCOLS

- **NTP (UDP 123):** syncs clocks across devices. Accurate time is essential for logs, Kerberos authentication, and certificates. Stratum 0 is the reference clock (atomic or GPS); lower stratum

= closer to the source.

- **PTP (Precision Time Protocol)**: sub-microsecond accuracy, used in finance, telecom, and industrial systems.
- **NTS (Network Time Security)**: adds authentication to NTP so attackers can't feed you false time.

3.5 Network Access and Management Methods

- **Site-to-site VPN**: connects entire networks (e.g., a branch to HQ) through VPN gateways. Always on, and users don't do anything.
- **Client-to-site (remote access) VPN**: an individual user connects to the corporate network with VPN client software. **Clientless** VPN runs in a web browser over TLS.
- **Split tunnel**: only corporate-bound traffic goes through the VPN; internet traffic goes directly out. Saves bandwidth but less control. **Full tunnel**: *all* traffic goes through the VPN, so it can be inspected. More secure but heavier.
- **Connection methods**: **SSH** (secure command line), **GUI** (web interface over HTTPS), **API** (automation), and **console** (direct serial cable to the device, works even when the network is down).
- **Jump box / jump host**: a hardened server admins connect to first, then reach sensitive devices from there. Creates one controlled, audited entry point.
- **In-band management**: managing devices over the same network that carries user traffic.
Out-of-band (OOB): a separate, dedicated management path (console servers, cellular modems) that still works when the production network is down.

04

Network Security

DOMAIN 4 • 14% OF THE EXAM

The smallest domain, but security concepts also show up in questions across the other domains. You already have a head start here from your cybersecurity coursework.

4.1 Basic Network Security Concepts

CORE TERMINOLOGY

- **CIA triad:** **Confidentiality** (only authorized people see data: encryption, access control), **Integrity** (data isn't altered: hashing, digital signatures), and **Availability** (systems are up when needed: redundancy, backups, DDoS protection).
- **Vulnerability:** a weakness. **Threat:** something that could exploit it. **Exploit:** the actual method or code that takes advantage of it. **Risk:** the likelihood and impact of a threat exploiting a vulnerability.

LOGICAL SECURITY

- **Encryption of data in transit** (TLS, IPsec, SSH) and **data at rest** (full-disk encryption, encrypted databases).
- **Certificates and PKI:** a Certificate Authority (CA) signs certificates that bind a public key to an identity. **Self-signed** certificates aren't signed by a trusted CA, so browsers warn about them. Fine for labs, not for public services.

IDENTITY AND ACCESS MANAGEMENT (IAM)

TERM	WHAT IT IS
MFA	Two or more factor types: something you know (password), have (phone, token), are (biometric). Also somewhere you are (location) and something you do.
SSO	Single Sign-On: log in once, access many systems.
RADIUS	AAA protocol for network access (Wi-Fi, VPN, 802.1X). UDP. Encrypts only the password.
TACACS+	Cisco AAA protocol, mostly for device administration. TCP 49. Encrypts the whole payload and separates authentication, authorization, and accounting.
LDAP	Queries directory services like Active Directory for user and group info.
SAML	XML-based standard for exchanging authentication between an identity provider and web apps. Common for web SSO.
Time-based authentication	Access allowed only during certain hours, or one-time passwords (TOTP) that expire after ~30 seconds.
Least privilege	Users get only the access they need to do their job, nothing more.
RBAC	Role-Based Access Control: permissions are assigned to roles (e.g., “HR”), and users get the role.

REMEMBER

AAA = **Authentication** (who are you?), **Authorization** (what can you do?), **Accounting** (what did you do?).

OTHER CONCEPTS

- **Geofencing**: allows or blocks access based on a device’s physical location.
- **Physical security**: cameras, locks, badge readers, access control vestibules (mantraps).
- **Deception technologies**: a **honeypot** is a fake system that lures attackers so you can detect and study them. A **honeynet** is a network of honeypots.
- **Compliance**: **PCI DSS** (rules for handling payment card data), **GDPR** (EU personal data privacy). **Data locality** laws require certain data to be stored within a specific country.
- **Network segmentation**: put risky or special-purpose devices on their own isolated segments: **IoT and IIoT** devices, **SCADA / ICS / OT** (industrial control systems), **guest** networks, and **BYOD** (personal devices). Limits how far an attacker can move.

4.2 Types of Attacks

ATTACK	HOW IT WORKS	DEFENSE
DoS / DDoS	Floods a target with traffic or requests until it can't serve legitimate users. DDoS uses many sources (often a botnet).	Rate limiting, upstream scrubbing services, CDNs, redundancy.
VLAN hopping	Attacker reaches a VLAN they shouldn't. Switch spoofing : pretends to be a switch to form a trunk. Double tagging : adds two 802.1Q tags, exploiting the native VLAN.	Disable auto-trunking (DTP), set access ports to access mode, change the native VLAN from VLAN 1.
MAC flooding	Fills the switch's MAC table with fake addresses until it fails open and floods all traffic to every port, like a hub.	Port security (limit MACs per port).
ARP spoofing / poisoning	Sends fake ARP replies so victims map the gateway's IP to the attacker's MAC. Enables on-path attacks.	Dynamic ARP Inspection (DAI), static ARP entries.
DNS spoofing / poisoning	Plants false DNS records in a cache so users get sent to malicious sites.	DNSSEC, secure resolvers.
Rogue DHCP server	Unauthorized DHCP server hands out a bad gateway or DNS server, redirecting traffic.	DHCP snooping (only trusted ports can send DHCP offers).
Rogue AP	Unauthorized AP plugged into the network, creating a backdoor.	Wireless IDS scans, 802.1X on switch ports.
Evil twin	Rogue AP with the <i>same</i> SSID as a legitimate network to trick users into connecting.	WPA3-Enterprise with certificate validation, user awareness.
On-path (man-in-the-middle)	Attacker secretly intercepts and possibly alters traffic between two parties.	Encryption (TLS, VPN), certificate validation.

SOCIAL ENGINEERING AND MALWARE

- **Phishing**: fraudulent messages that trick users into giving up credentials or running malware. Variants: spear phishing (targeted), whaling (executives), vishing (voice), smishing (SMS).
- **Dumpster diving**: searching trash for sensitive info. Defense: shred documents.
- **Shoulder surfing**: watching someone enter a password. Defense: privacy screens, awareness.
- **Tailgating**: following an authorized person through a secure door. (**Piggybacking** is the same thing but with the person's consent.) Defense: access control vestibules, guards, training.
- **Malware**: viruses, worms (self-spreading across the network), trojans, ransomware, spyware, rootkits.

4.3 Network Security Features, Defense Techniques, and Solutions

- **Device hardening:** disable unused ports and services, change default passwords, use secure management protocols (SSH, not Telnet), keep firmware patched.
- **NAC (Network Access Control):** checks devices before letting them on the network. **802.1X** is port-based authentication with three roles: the **supplicant** (client), the **authenticator** (switch or AP), and the **authentication server** (RADIUS).
- **Port security:** limits which or how many MAC addresses can use a switch port. Violations can shut down the port (err-disabled).
- **MAC filtering:** allow or deny by MAC address. Weak on its own, because MACs are easy to spoof.
- **Key management:** securely generating, distributing, storing, rotating, and revoking encryption keys.
- **ACLs (Access Control Lists):** ordered permit/deny rules on routers and firewalls. Processed **top-down, first match wins**, and there's an **implicit deny** at the end. Put specific rules above general ones.
- **URL filtering** blocks websites by address or category. **Content filtering** inspects what's actually in the traffic.
- **Security zones:** **trusted** (internal), **untrusted** (internet), and the **screened subnet** (formerly called the DMZ), which holds public-facing servers like web and mail, isolated from the internal network.

REMEMBER

ACL questions are common in PBQs. Read the rules top to bottom and stop at the first match. If nothing matches, the traffic is denied.

05

Network Troubleshooting

DOMAIN 5 • 24% OF THE EXAM

This is the biggest domain, and it draws on everything else. Most questions are scenarios: a user reports a symptom, and you pick the cause, the next step, or the right tool.

5.1 The Troubleshooting Methodology

Memorize these seven steps **in order**. CompTIA loves “what should the technician do NEXT?” questions.

STEP	WHAT TO DO
1. Identify the problem	Gather information, question users, identify symptoms, determine if anything changed, duplicate the problem if possible, approach multiple problems individually.
2. Establish a theory of probable cause	Question the obvious. Consider multiple approaches: top-to-bottom or bottom-to-top through the OSI model, or divide and conquer.
3. Test the theory to determine the cause	If confirmed, determine the next steps to fix it. If not confirmed, establish a new theory or escalate.
4. Establish a plan of action	Plan the fix and identify potential effects (downtime, who's affected, change management).
5. Implement the solution or escalate	Make the fix, or escalate if it's beyond your authority or skill.
6. Verify full system functionality	Confirm it works, and implement preventive measures if applicable.
7. Document findings, actions, outcomes, and lessons learned	Always last. Helps the next person and builds a knowledge base.

REMEMBER

Mnemonic: **I Eat Tacos, Eat Ice cream, Very Delicious** → Identify, Establish theory, Test, Establish plan, Implement, Verify, Document.

5.2 Cabling and Physical Interface Issues

CABLE PROBLEMS

- **Incorrect cable type:** single-mode vs. multimode fiber mismatch, wrong copper category for the speed (e.g., Cat 5 for 1 Gbps), or UTP where STP is needed near heavy EMI.
- **Attenuation:** the signal weakens over distance. Copper Ethernet's limit is **100 meters**. Too long = errors and dropped links.
- **Interference (EMI):** from motors, fluorescent lights, and power cables. Fix with shielded cable, rerouting, or fiber.
- **Crosstalk:** signal bleeding between wire pairs, often from untwisting too much of the pair at the termination. **NEXT** = near-end, **FEXT** = far-end.
- **Improper termination:** bad crimp or wrong pin order. Test with a cable tester.
- **TX/RX transposed:** transmit and receive are reversed, common with fiber patch cables. The link won't come up. Swap the pair.

INTERFACE COUNTERS AND STATUS

COUNTER / STATUS	WHAT IT MEANS	LIKELY CAUSE
CRC errors	Frames arrived corrupted (failed the checksum).	Bad cable, EMI, duplex mismatch.
Runts	Frames smaller than 64 bytes.	Collisions, duplex mismatch, bad NIC.
Giants	Frames larger than the MTU (1518 bytes).	MTU / jumbo frame mismatch, bad NIC.
Drops / discards	Frames thrown away.	Congestion, full buffers, QoS policy.
Error-disabled	Switch shut the port down automatically.	Port security violation, BPDU guard, loop detected.
Administratively down	An admin manually disabled the port ("shutdown").	Fix: "no shutdown".
Suspended	Port suspended from a link aggregation group.	LACP misconfiguration or mismatch.

HARDWARE ISSUES

- **PoE problems:** the switch's total **power budget exceeded** (devices at the end of the list don't power on), or an **incorrect standard** (device needs more power than the port supplies).
- **Transceiver problems:** mismatched transceivers (wrong speed, wavelength, or fiber type) or low **signal strength** (dirty connectors, too long a run, bad fiber).

POE STANDARD	NAME	POWER AT THE SWITCH PORT
802.3af	PoE	15.4 W
802.3at	PoE+	30 W
802.3bt Type 3	PoE++	60 W
802.3bt Type 4	PoE++	~90–100 W

5.3 Network Service Issues

- **Spanning tree problems: network loops** cause broadcast storms, maxed-out CPU, and flapping MAC tables. A poorly chosen **root bridge** sends traffic through a slow path. Check port roles and states.
- **Incorrect VLAN assignment:** the host can reach devices on the wrong subnet but not its own. Or it gets an IP from the wrong DHCP scope.
- **ACL misconfiguration:** a rule blocks legitimate traffic. Remember the implicit deny at the end.
- **Routing problems:** missing or wrong routes in the **routing table**, or a missing **default route**, so traffic to unknown networks gets dropped.
- **Address pool exhaustion:** the DHCP scope ran out, so new clients get APIPA (169.254.x.x) addresses. Fix: expand the scope or shorten the lease time.
- **Incorrect default gateway:** the host can reach its local subnet but nothing beyond it.
- **Duplicate IP address:** two devices with the same IP. Causes intermittent connectivity and “IP conflict” warnings.
- **Incorrect subnet mask:** the host miscalculates which addresses are local, so it can reach some devices but not others.

REMEMBER

Can ping its own subnet but not the internet → check the **default gateway**. Can ping by IP but not by name → check **DNS**. Has a 169.254 address → check **DHCP**.

5.4 Performance Issues

- **Congestion / contention:** more traffic than the link can carry. **Bottleneck:** the slowest point in the path limits everything.
- **Bandwidth** is the theoretical maximum capacity. **Throughput** is what you actually get, which is always lower.
- **Latency:** the delay for data to travel from source to destination (measured in ms).
- **Jitter:** *variation* in latency between packets. Ruins voice and video calls (choppy audio). Fix with QoS.
- **Packet loss:** packets that never arrive. Causes retransmissions (TCP) or gaps (UDP voice/video).

WIRELESS PERFORMANCE ISSUES

- **Interference / channel overlap:** nearby APs on overlapping channels (e.g., channels 1 and 3 on 2.4 GHz). Fix: use 1, 6, 11, or move to 5/6 GHz.
- **Signal degradation or loss:** distance, walls, metal, water (including people). Check the RSSI (signal strength); around -67 dBm or better is good for voice.
- **Insufficient coverage:** dead zones. Fix: a site survey and more or better-placed APs.
- **Client disassociation:** clients keep dropping off. Could be interference, weak signal, or a deauthentication attack.
- **Roaming misconfiguration:** clients stick to a far AP instead of moving to a closer one, or SSIDs and security settings don't match across APs.

5.5 Tools and Protocols for Troubleshooting

COMMAND-LINE TOOLS

COMMAND	USE
ping	Tests reachability using ICMP echo. Start here.
tracert (Linux) / tracert (Windows)	Shows each router hop to a destination. Finds where traffic stops.
ipconfig (Windows) / ip, ifconfig (Linux)	Shows the host's IP config. ipconfig /release and /renew refresh DHCP. ipconfig /flushdns clears the DNS cache.
nslookup / dig	Queries DNS to test name resolution. dig is the more detailed Linux tool.
netstat	Shows active connections and listening ports. Useful for spotting unexpected services.
arp -a	Shows the IP-to-MAC mappings the host has learned.
tcpdump	Command-line packet capture (Wireshark is the GUI equivalent).
nmap	Port and network scanner. Finds hosts, open ports, and running services.
pathping (Windows)	Combines ping and traceroute, showing packet loss at each hop.

DEVICE COMMANDS (CISCO-STYLE)

COMMAND	SHOWS
show interface	Interface status, speed, duplex, and error counters (CRC, runs, giants).
show ip route / show route	The routing table.
show mac-address-table	Which MAC addresses were learned on which ports.
show vlan	VLANs and their assigned ports.
show arp	The device's ARP table.
show running-config / show config	The current configuration.
show power inline	PoE power usage per port and remaining budget.

OTHER SOFTWARE TOOLS AND PROTOCOLS

- **Protocol analyzer** (e.g., Wireshark): captures and decodes packets for deep inspection.
- **Speed tester**: measures actual throughput to compare against expected bandwidth.
- **LLDP / CDP**: discovery protocols where devices advertise their identity to neighbors. Shows which switch and port a device is plugged into. LLDP is the open standard; CDP is Cisco's.

HARDWARE TOOLS

TOOL	USE
Toner and probe	Traces an unlabeled cable to find where it ends up (e.g., in a patch panel).
Cable tester	Checks continuity, wire map, shorts, and crossed pairs. Certifiers also verify the cable meets category specs.
Network tap	Hardware device that copies all traffic on a link for monitoring. More reliable than port mirroring.
Wi-Fi analyzer	Shows nearby SSIDs, channels, and signal strength. Used to find interference and choose channels.
Visual fault locator	Shines a bright visible laser into fiber to find breaks or sharp bends.
OTDR	Optical Time-Domain Reflectometer: measures fiber length and locates faults precisely.
Loopback adapter	Plugs into a port to test whether the port itself works.

Practice Questions

32 QUESTIONS · ANSWER KEY FOLLOWS

- 01 A user's laptop has an IP address of 169.254.33.12. What is the MOST likely cause?**
- A Duplicate IP address
 - B The laptop can't reach a DHCP server
 - C Incorrect DNS server
 - D Wrong default gateway
- 02 At which OSI layer does a switch forward frames using MAC addresses?**
- A Layer 1
 - B Layer 2
 - C Layer 3
 - D Layer 4
- 03 Which port must be opened on a firewall to allow secure remote command-line management of a switch?**
- A 22
 - B 23
 - C 443
 - D 3389
- 04 How many usable host addresses are available in a /27 subnet?**
- A 14
 - B 30
 - C 32
 - D 62
- 05 What is the broadcast address of 10.1.1.130/26?**
- A 10.1.1.127
 - B 10.1.1.159
 - C 10.1.1.191
 - D 10.1.1.255
- 06 A router has routes to 172.16.0.0/16 learned via OSPF and 172.16.5.0/24 learned via a static route. Which route is used for traffic to 172.16.5.10?**
- A The OSPF route, because it was learned dynamically
 - B The static route, because it has the longest prefix match
 - C Both, load balanced
 - D Neither, the traffic is dropped

- 07 Which protocol allows two routers to share a virtual IP address so hosts keep a working default gateway if one router fails?**
- A STP
 - B LACP
 - C VRRP
 - D BGP
- 08 A network has redundant links between switches, and suddenly experiences a broadcast storm. Which protocol is MOST likely misconfigured or disabled?**
- A STP
 - B DHCP
 - C OSPF
 - D 802.1X
- 09 Which standard adds VLAN tags to Ethernet frames on a trunk link?**
- A 802.1X
 - B 802.1Q
 - C 802.3af
 - D 802.11ac
- 10 Which three channels should be used on 2.4 GHz to avoid overlap in North America?**
- A 1, 5, 9
 - B 1, 6, 11
 - C 2, 7, 12
 - D 36, 40, 44
- 11 A company wants each employee to authenticate to Wi-Fi with their own credentials. Which should be configured?**
- A WPA2-Personal
 - B WPA3-Personal with SAE
 - C WPA3-Enterprise with RADIUS
 - D An open network with a captive portal
- 12 How tall is a 1U piece of rack equipment?**
- A 1 inch
 - B 1.5 inches
 - C 1.75 inches
 - D 2 inches

- 13 A company can afford to lose no more than 15 minutes of data in a disaster. Which metric does this describe?**
- A RTO
 - B RPO
 - C MTTR
 - D MTBF
- 14 Which DR site has space and power but no equipment or current data?**
- A Hot site
 - B Warm site
 - C Cold site
 - D Mirror site
- 15 Clients on a new subnet are not receiving IP addresses. The DHCP server is on a different subnet and is working for other clients. What is MOST likely missing?**
- A A DNS PTR record
 - B A DHCP relay (IP helper) on the router
 - C A default route on the DHCP server
 - D An NTP server
- 16 Which DNS record type maps a hostname to an IPv6 address?**
- A A
 - B AAAA
 - C PTR
 - D CNAME
- 17 Which SNMP version should be used to provide both authentication and encryption?**
- A SNMPv1
 - B SNMPv2c
 - C SNMPv3
 - D All versions support encryption
- 18 Which syslog severity level is the MOST severe?**
- A 0 – Emergency
 - B 3 – Error
 - C 5 – Notice
 - D 7 – Debug

- 19 A remote worker's VPN sends only corporate traffic through the tunnel, while streaming and web traffic go directly to the internet. What is this called?
- A Full tunnel
 - B Split tunnel
 - C Site-to-site VPN
 - D Clientless VPN
- 20 An administrator needs to manage a router when the production network is completely down. Which method should be used?
- A In-band management over SSH
 - B Out-of-band management via a console server
 - C HTTPS web GUI
 - D SNMP polling
- 21 An attacker sets up an access point with the same SSID as the company's Wi-Fi to capture credentials. What attack is this?
- A Rogue DHCP
 - B MAC flooding
 - C Evil twin
 - D VLAN hopping
- 22 Which switch feature prevents an attacker from overflowing the MAC address table?
- A Port security
 - B DHCP snooping
 - C BPDU guard
 - D Jumbo frames
- 23 In 802.1X, which device acts as the authenticator?
- A The client laptop
 - B The RADIUS server
 - C The switch or access point
 - D The domain controller
- 24 A web server that must be reachable from the internet should be placed in which zone?
- A Trusted internal network
 - B Screened subnet
 - C Guest VLAN
 - D Management VLAN

- 25 Which AAA protocol encrypts the entire payload and is commonly used for network device administration?**
- A RADIUS
 - B TACACS+
 - C LDAP
 - D SAML
- 26 A technician has identified the problem and established a theory of probable cause. What should the technician do NEXT?**
- A Implement the solution
 - B Document findings
 - C Test the theory to determine the cause
 - D Establish a plan of action
- 27 A user can ping 8.8.8.8 but cannot browse to websites by name. What is the MOST likely issue?**
- A Incorrect default gateway
 - B DNS failure
 - C Bad cable
 - D DHCP scope exhausted
- 28 A switch interface shows a high number of CRC errors and late collisions. Which is the MOST likely cause?**
- A Duplex mismatch
 - B DNS misconfiguration
 - C Incorrect VLAN
 - D Expired DHCP lease
- 29 VoIP calls are choppy, and monitoring shows highly variable delay between packets. What is this called?**
- A Attenuation
 - B Jitter
 - C Crosstalk
 - D Bandwidth
- 30 Several new PoE cameras will not power on, but the first cameras connected work fine. What is the MOST likely cause?**
- A Wrong VLAN
 - B The switch's PoE power budget was exceeded
 - C Duplicate IP addresses
 - D STP blocking the ports

- 31 A technician needs to find which patch panel port an unlabeled wall jack connects to. Which tool should be used?**
- A OTDR
 - B Toner and probe
 - C Wi-Fi analyzer
 - D Protocol analyzer
- 32 Which command shows the hops a packet takes to reach a destination on a Windows PC?**
- A ping
 - B tracert
 - C nslookup
 - D netstat

Answer Key

#	ANSWER	WHY
01	B	169.254.x.x is an APIPA address, which a host assigns itself when DHCP fails. (1.7, 5.3)
02	B	Switches are Layer 2 devices. Frames and MAC addresses live at the Data Link layer. (1.1)
03	A	SSH uses TCP 22. Telnet (23) is insecure, 443 is HTTPS, and 3389 is RDP. (1.4)
04	B	/27 leaves 5 host bits: $2^5 - 2 = 30$. (1.7)
05	C	/26 has a block size of 64. Blocks start at 0, 64, 128, 192, so 130 is in 128–191, and the broadcast is 10.1.1.191. (1.7)
06	B	Longest prefix match is checked first, before administrative distance. The /24 is more specific than the /16. (2.1)
07	C	VRRP (like HSRP) is a First Hop Redundancy Protocol. (2.1)
08	A	Spanning Tree Protocol prevents Layer 2 loops, which cause broadcast storms. (2.2, 5.3)
09	B	802.1Q is VLAN tagging. 802.1X is port authentication, 802.3af is PoE, and 802.11ac is Wi-Fi 5. (2.2)
10	B	Channels 1, 6, and 11 are the only non-overlapping 2.4 GHz channels in North America. 36/40/44 are 5 GHz channels. (2.3)
11	C	Enterprise mode uses 802.1X and a RADIUS server for individual user authentication. Personal modes use one shared key. (2.3)
12	C	One rack unit is 1.75 inches. (2.4)
13	B	RPO is the maximum acceptable data loss, measured in time. RTO is acceptable downtime. (3.3)
14	C	A cold site is the cheapest and slowest to bring online. (3.3)
15	B	DHCP Discover is a broadcast, which routers don't forward. A relay agent forwards it to the remote server. (3.4)
16	B	AAAA records are for IPv6. A records are for IPv4. (3.4)
17	C	Only SNMPv3 supports authentication and encryption. v1/v2c use plaintext community strings. (3.2)
18	A	Lower numbers are more severe, and 0 is Emergency. (3.2)
19	B	Split tunnel sends only corporate-bound traffic through the VPN. (3.5)
20	B	Out-of-band management uses a separate path that works even when the production network fails. (3.5)
21	C	An evil twin impersonates a legitimate network's SSID. (4.2)

#	ANSWER	WHY
22	A	Port security limits the number of MAC addresses on a port, which stops MAC flooding. (4.2, 4.3)
23	C	Supplicant = client, authenticator = switch/AP, authentication server = RADIUS. (4.3)
24	B	The screened subnet (DMZ) holds public-facing services, isolated from the internal network. (4.3)
25	B	TACACS+ uses TCP 49 and encrypts the whole payload. RADIUS encrypts only the password. (4.1)
26	C	Step 3 of the methodology is testing the theory. (5.1)
27	B	Reaching an IP but not names points to DNS. A gateway or cable problem would also break the ping. (5.3)
28	A	A duplex mismatch (or bad cabling) produces CRC errors and late collisions. (5.2)
29	B	Jitter is variation in latency. QoS helps fix it. (5.4)
30	B	Once the total PoE budget is used up, additional devices don't get power. (5.2)
31	B	A toner and probe traces copper cables to their other end. (5.5)
32	B	tracert (traceroute on Linux) lists each router hop. (5.5)